



D H Y U T I

AI FOR SCIENCE AND ENGINEERING

CAISc 2026 Pre-Conference Workshop

# DhyutimaanPI

*A skill-based agentic framework for PINN research*



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# Agenda

01

## The Problem

Motivation, Automating thesis work, and scaling research

02

## The Four Agents

Surveyor, Framer, Implementer, Analyst

03

## Failure Modes and Guard rails

What the framework should refuse to compromise on?

04

## Worked Examples and Workflows to run

2D heat conduction

05

## Where to Run It

Claude Code, Claude.ai, Cursor, Antigravity

06

## Discussion & Next Steps

Recommended open problems



# 01

SECTION 01 / 06

## The Problem

Background, Motivation and Scaling the research methodology

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*Can I have agents reproduce my thesis? Can my thesis become a living, evolving body of work?*



# How it started?

“Can my thesis transform into an evolving body of work?”

Why should the thesis not evolve as new knowledge adds?

How do I scale such evolution?

How do I automate such work with AI?

What methodology guard rails should I adopt?



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**Data-driven and physics-informed deep  
learning for surrogate modeling of  
unsteady flows past moving bodies**



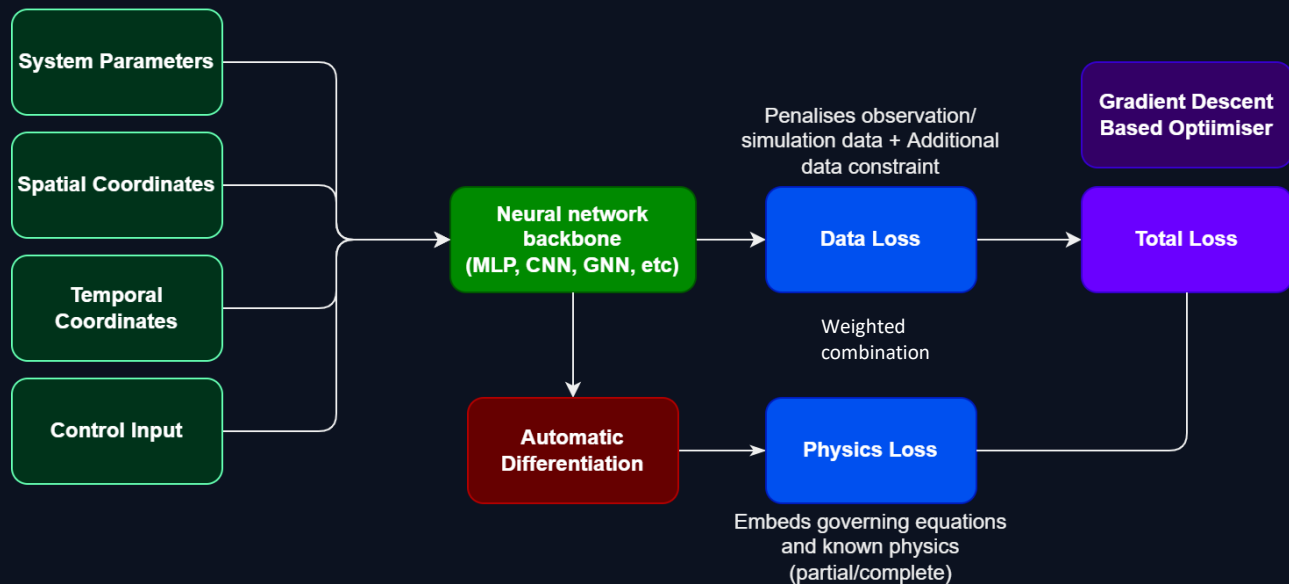
*A Thesis*

*Submitted by*

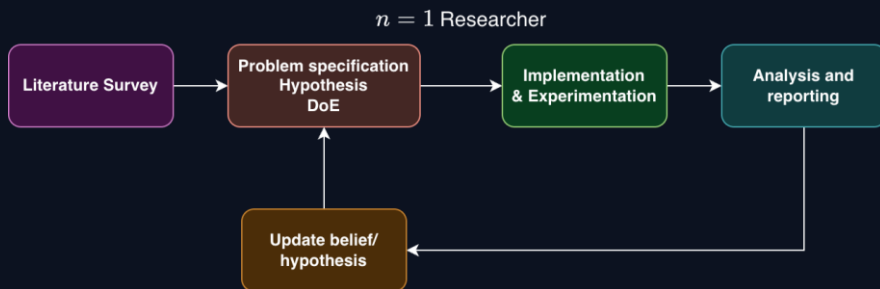
**RAHUL SUNDAR**

# Physics informed neural network (PINN) – Quick overview

(Raissi et. al (2019))



# Standard research methodology



## AN AVERAGE HUMAN RESEARCHER

“Every step by hand, every time”

### Informal handoffs

Literature survey, problem framing, code development, analysis & reporting live across different modes and not integrated — nothing is reusable, nothing is comparable across labs.

### Celebratory analysis

Loss went down? Ship it. Boundary error 70× the interior? Physics loss residual goes unflagged. Relative explainability of variants not studied.

### Slow and inconsistent

Costly to scale, hard to reproduce; junior researchers reinvent practice on every new problem.

# 02

SECTION 02 / 06

## The Four Agents

What each one knows, what each one refuses to do

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*Topics › Surveyor · Framer · Implementer · Analyst*



# Our approach: a chain of four specialized agents

1

Surveyor

*literature-survey-pinn*

Literature survey

Produces a structured PINN-specific knowledge base.

Key - **Taxonomy** of variants, training tricks, failure modes — operational, not encyclopedic.

2

Framer

*pinn-problem-spec*

Problem specification,  
Hypothesis and DoE

Turns a vague idea into a **falsifiable/verifiable problem-spec.md**.

**Key – Query expansion!**  
Mandatory reference solution, quantitative hypotheses, anticipated failure modes.

3

Implementer

*pinn-scaffold*

Scaffolds the code, and sets up experiments

Generates Numpy / PyTorch + torch.func implementation with a **verification harness** that compares against the spec's reference solution.

4

Analyst

*pinn-analysis-report*

Analyses, audits, critics, and reviews

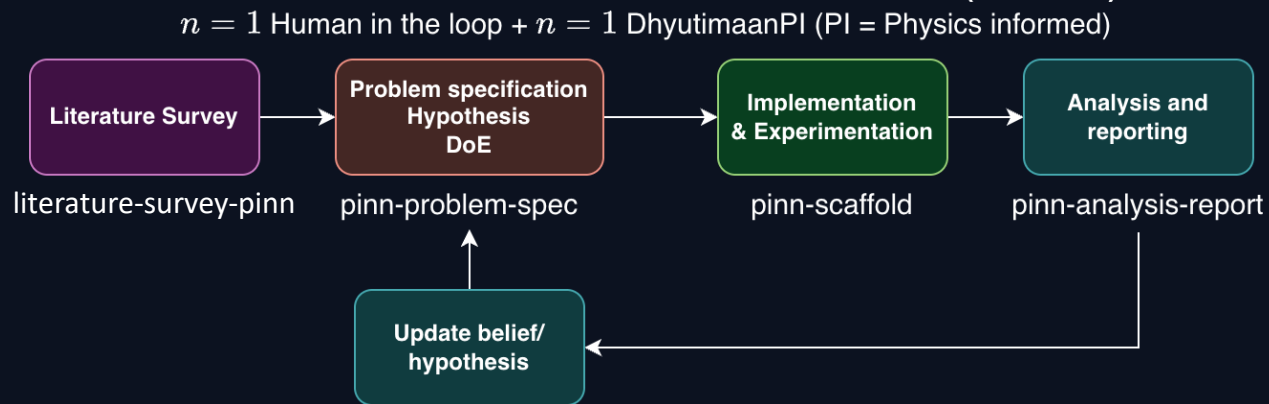
**Suspicious by default.** Audits every failure mode, marks single-seed wins as “inconclusive”, writes a **report that reads like a reviewer**.

Each agent is a self-contained SKILL.md file. Handoffs are plain Markdown / Python / JSON artifacts — readable, editable, version-controllable.



# DhyutimaanPI: a chain of four specialized agents

*Each agent is a self-contained SKILL.md file. Handoffs are plain Markdown / Python / JSON artifacts — readable, editable, version-controllable.*



Scan for repo



*Let's deep dive into the Skill files!*



# Surveyor — reads the literature, builds a knowledge base

## ■ What it does

Conducts focused literature surveys on PINN topics. Searches arXiv, Semantic Scholar, OpenReview, GitHub — not just one source.

## ■ Query expansion

Expands along 4 axes: variants (vanilla, XPINN, causal), training tricks, failure modes, and PDE class. Inconsistent terminology demands it.

## ■ Output

A structured pinn-knowledge-base.md with a taxonomy and a “Recommended starting points” section that feeds directly into the next agent.

## ■ What it refuses

Encyclopedic dumps and unverified claims. Every assertion is tied to a paper, by author and year; PINN folklore is flagged as folklore.

pinn-knowledge-base.md

—

### # Taxonomy

- Vanilla PINN (Raissi 2019)
- Causal weighting (Wang 2022)
  - XPINN, FBPINN
- Fourier features, SIREN

### # Failure modes

- Spectral bias
- Propagation failure

# Recommended starts → Framer

*Output artifact: a structured Markdown knowledge base, taxonomy-organized.*

# Framer — writes a falsifiable problem-spec.md

## ■ What it does

Turns an informal “let’s try a PINN on X” into a 9-section problem spec: PDE, BCs, reference solution, hypotheses, success criteria, failure modes.

## ■ Falsifiable hypotheses

“The PINN will work” is rewritten to “rel.  $L^2$  error  $< 10^{-3}$  on a  $200 \times 200$  grid within 5k Adam steps.” Intervention + metric + threshold.

## ■ Reference is mandatory

Analytical  $\rightarrow$  method of manufactured solutions  $\rightarrow$  high-fidelity numerical. Refuses to proceed without one. PINN without a reference is a confident hallucination.

## ■ Anticipates failure

Required “known failure modes” section: BC drift, spectral bias, causality violation. Feeds the Analyst’s audit checklist downstream.

```
problem-spec.md
—
## Governing equation
— $\nabla \cdot (k \nabla T) = q$ 

## Reference (MMS)
 $T^* = \sin \pi x \sin \pi y$ 
 $q = 2\pi^2 \sin \pi x \sin \pi y$ 

## H1: rel.  $L^2 < 10^{-3}$ 
within 5k Adam steps

## Watch for
BC drift, spectral bias
```

*The spec is the contract: every downstream agent reads it as input.*

# Implementer — scaffolds verifiable PyTorch code

## ■ Default stack

PyTorch with torch.func (grad, vmap, jacrev, hessian)/ Numpy if discrete Physics loss. The PDE residual is ~10 lines of explicit Python — transparent, not hidden inside a higher-level library. DeepXDE available as alternate.

## ■ Verification harness

verify.py is non-negotiable. Computes relative  $L^2$  against the reference, evaluates each hypothesis from the spec, writes a structured JSON verdict the Analyst consumes.

## ■ Per-term logging

training\_log.csv keeps PDE, BC, and IC loss in separate columns. A low total loss can hide BC error 70× the residual; aggregation makes that invisible.

## ■ Apple Silicon ready

MPS backend on M-series Macs, with CPU fallback. Small PINNs often train fastest on CPU anyway — kernel-launch overhead dominates. Demo problems converge in < 60 seconds.

```
pinn_run/  
├─ problem.py  
  (pde_residual via torch.func.hessia/ np.diff)  
├─ model.py (MLP)  
├─ train.py (Adam → L-BFGS)  
├─ verify.py  
├─ verification.json  
└─ run.py
```

*Self-contained module: ~250 lines total, all human-readable.*

# Analyst — adversarial reader of the training output

## ■ Hypothesis verdicts

Marks each hypothesis confirmed / refuted / inconclusive. Single-seed wins default to inconclusive — one number is anecdote, not evidence.

## ■ Failure-mode audit

Walks every failure mode the spec listed: BC drift, spectral bias, causality violation. Each one is a computable signal, not folklore.

## ■ Honest limitations

Required section: single vs. multi-seed, reference quality, in-domain vs. out-of-domain. The section users skip and the one that matters most.

## ■ Concrete next steps

Not “train for longer.” Phrased as hypotheses for the next iteration: “Switch to causal weighting ( $\epsilon=100$ ); predict rel.  $L^2$  at  $t>0.7$  drops by 2x.”

analysis-report.md

—

# Headline

rel.  $L^2 = 7 \times 10^{-4}$  ✓

# Verdicts

H1: confirmed (single seed)

# Audit

⚠ BC loss / PDE loss = 70x  
raise  $\lambda_b$  from 1 to 10

# Limitations

single seed, MMS only

*A successful run that still gets flagged. Loss went down; the model still satisfied the equation in the wrong place.*

# 03

SECTION 03 / 06

## Failure modes and guardrails

Four commitments the framework should stick to!

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# Four commitments the framework refuses to compromise on

## ■ 01 — Query expansion and multi-axis – synonymous literature deep dive

A query when vague can mean multiple things, requires user intent confirmation. Thorough, deep dive into multiple axis of PINN literature such as variants, optimisers, training tricks, sampling approaches, architecture, etc becomes necessary.

## ■ 02 — Falsifiable/verifiable hypotheses, no exceptions

Every hypothesis must specify what changes, what is measured, and the quantitative threshold for confirmation. “The model works well” is rejected. Convergence of the loss is necessary but not sufficient.

## ■ 03 — Reference solution is mandatory

Analytical solution → method of manufactured solutions → high-fidelity numerical reference. A PINN without an error measurement cannot be distinguished from a confidently-wrong PINN.

## ■ 04 — Analysis is adversarial by design

The Analyst is built to be suspicious. It audits every failure mode the spec listed, flags single-seed claims as inconclusive, and reports loss imbalance even when the headline metric passes. Reads like a reviewer, not a vendor.

*Enforced by: literature-survey-pinn, pinn-problem-spec, pinn-scaffold, pinn-analysis-report*



# 04

SECTION 04 / 06

## Examples and Workflows

2D Heat conduction example with analytical references and DhyutimaanPI workflows

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*Topics › 2D heat conduction · MMS as reference · Parallel DhyutimaanPI workflow*





# Example: 2D steady-state heat conduction (Claude code – Serial workflow)

## Physical setup

Square plate  $\Omega = [0,1]^2$ . Isotropic conductivity  $k=1$ . Homogeneous Dirichlet BCs. Distributed source  $q(x,y)$  chosen so the solution is known exactly.

## Manufactured solution

Pick  $T^*(x,y) = \sin(\pi x) \cdot \sin(\pi y)$ .  
Substitute into  $-\Delta T = q$  to derive  $q = 2\pi^2 \sin(\pi x) \cdot \sin(\pi y)$ .  
Network output is compared directly against  $T^*$ .

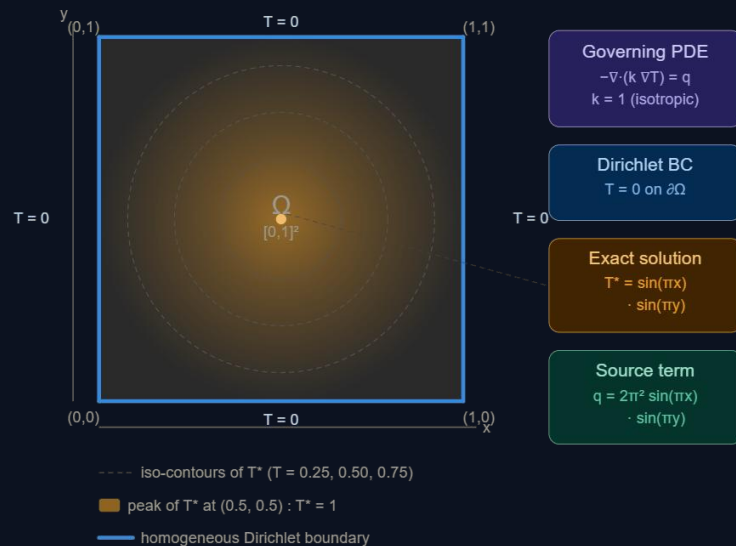
## Hypotheses

**H1: Error drop** - rel.  $L_2$  Error  $< 10^{-3}$  in 5k Adam steps.  
**H2: Optimiser** - L-BFGS reduces error  $10\times$  further.  
**H3: Loss weight** -  $\lambda_b = 10$  cuts BC error  $5\times$ .

## Why this benchmark

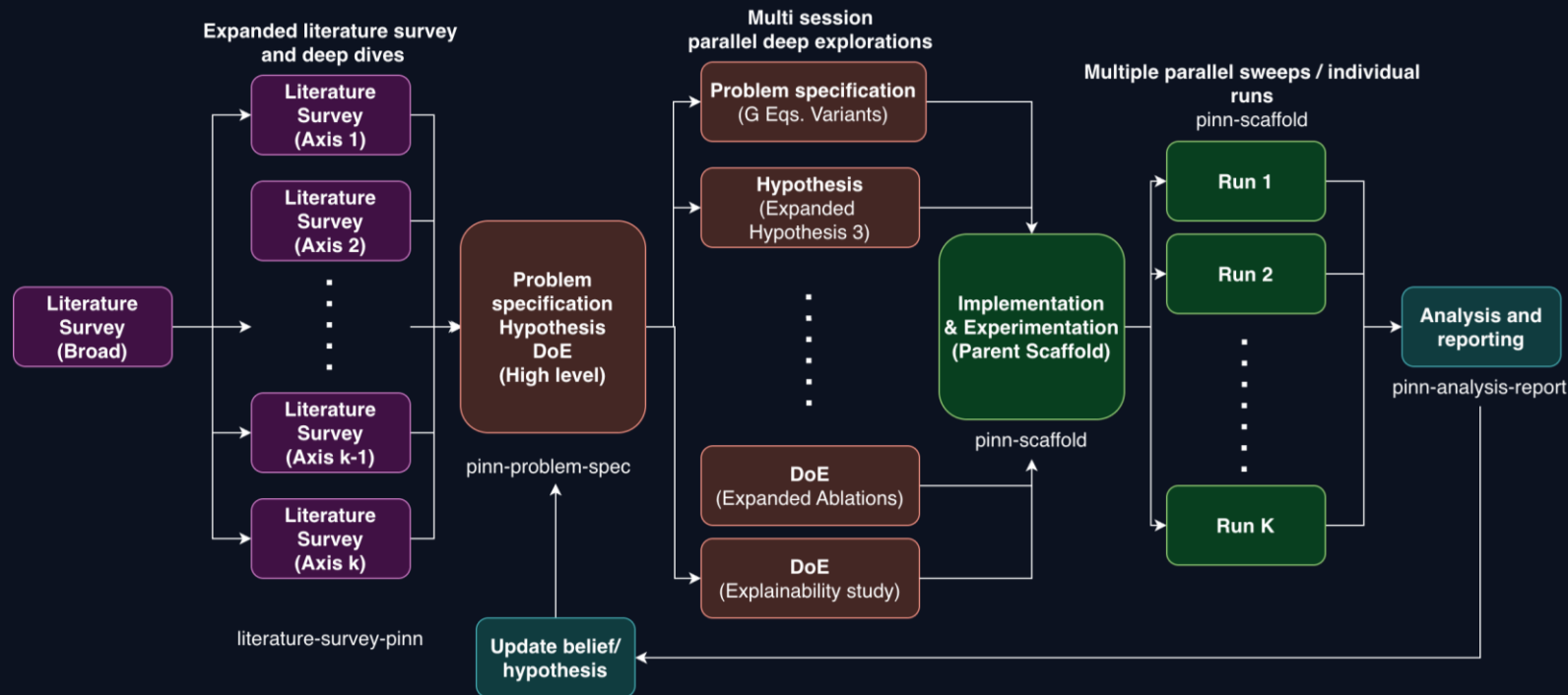
Smallest non-trivial benchmark: scalar, steady, analytical reference, converges in  $< 25$  min on a laptop. Perfect first contact.

Reference: *method of manufactured solutions* (Roache 2002)



*2D Poisson with manufactured source. The full spec ships in the workshop repo.*

# How to scale using DhyutmaanPI ? – Parallel workflow



# 05

SECTION 05 / 06

## Where to run it?

Claude, IDEs, etc.

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## Where to run it — same skills, four environments

1

### Claude Code/Cowork

#### Primary workflow

Skills auto-load from `.claude/skills/`. Best for end-to-end runs — agent uses your local terminal, your files, your Python. Recommended for the workshop hands-on.

2

### Claude.ai

#### Browser-based

Upload skills as `.skill` packages (Pro+) or drop into a Project. Useful for survey and analysis stages; train locally and upload outputs back.

3

### Cursor

#### IDE-integrated

Drop skills in `.cursor/skills/` or rely on `AGENTS.md` cross-tool reading. Pin Claude as the model. Composer handles multi-file scaffold output well.

4

### Antigravity

#### Multi-agent native

Closest match to a research pipeline. Skills in `.agents/skills/` + `AGENTS.md` defining the four-agent team. Agent Manager runs the full chain with review at each handoff.

Same `SKILL.md` files in every environment. Setup guides in docs/ for each. `AGENTS.md` works cross-tool — install once, run anywhere.

# 06

SECTION 05 / 06

## Discussion, Q&A, and open problems

Recommended open problems to solve using DhyutimaanPI

*P1: Forward problems – Rapidly solving 2D Unsteady Navier Stokes (External flows past largely deforming bodies)*

*P2: Inverse Problems - 3D flow reconstruction from 2D planar flows*



Interest form





## DhyutimaanPI core:

- PINNs have multiple failure modes - Literature driven hypothesis design, DoE, and error measurement are crucial!
- What's measurable can be mostly controllable.
- Mandatory reference solutions!

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*This is the single sentence that the rest of the framework defends. The agents, the artifacts, the audit checklists — all exist to make sure you never publish a PINN result you cannot measure.*



## References & Further Reading

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# Thank You

*Questions, discussion, and feedback are welcome.*

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